

Dhruva Bhat

858-205-4029 | dhruva.betkoppa@gmail.com | [linkedin.com/in/dhruvabhat](https://www.linkedin.com/in/dhruvabhat)

EDUCATION

University of California, Berkeley

Berkeley, CA

BS in Electrical Engineering and Computer Science (EECS) and Bioengineering

Expected Graduation: May 2028

PROFESSIONAL EXPERIENCE

Oak Ridge National Lab

May 2026 - July 2026

AI Platform Engineering Intern

- Engineered an asynchronous AI pipeline using FastAPI to orchestrate Google CoScientist, multi-agent tournament reasoning, retrieval-augmented generation (RAG), and molecular diffusion models for candidate generation
- Implemented product stability for modular API services with asynchronous task execution, retry mechanisms, fallback strategies, caching layers, and service orchestration to ensure robustness across multistage workflows
- Containerized services with Docker to standardize dependency management and enable multi-environment usage

Blueprint at Berkeley

Feb 2026 - Present

Full-Stack and Algorithms Developer

- Built and deployed a cloud-based volunteer application for Amigos de Los Rios using React, Supabase, Next.js, and AWS Lambda implementing authentication workflows and dynamic operational dashboards for admin
- Designed a geospatial routing algorithm integrating REST API data with PostgreSQL from PlanItGeo to generate volunteer routes based on time, location, and resource constraints, through time-based waterfall prioritization

Carnegie Mellon University

March 2025 - Present

AI Safety Research Assistant

- Developed a modular backend pipeline for orchestrating LLM experiments across multiple model environments, integrating inference APIs, experiment configuration, and automated data analysis
- Collaborating with Professor Sarah H. Cen researching bounded rationality of LLMs in institutional settings through benchmarking model behavior against game-theory baselines

University of California, San Francisco

Aug 2025 - Present

Machine Learning/Bioinformatics Research Assistant

- Implemented an unsupervised machine learning pipeline in R to analyze genomic data, including data cleaning, feature selection, hierarchical clustering, PCA, network-based outlier detection, and batch-effect correction
- Ran pipeline on 125,000+ patient data rows, analyzing gene marker quantity in comparison to cell type prevalence across tissue samples, working with Professor Oldham to identify optimal lung cancer gene markers
- Engineering data scraper and pipelining to gather and process GRE data prior to analysis; developing neural network model to compile analysis across studies to highlight optimal gene markers for specified cell types

Garcia Center for Polymers at Engineering Interfaces

Jun 2024 - Dec 2024

Researcher at Stony Brook University

- Thin-Film Research: Used 4D manifold learning in PyTorch, developed a Gaussian Process Regression model correlating molecular weight and film-thickness predicting commercial polystyrene semiconductor wafer thickness
- Hydrogel Research: Conducted rheological analysis of F88DMA cross-linked with APS and TEMED; applied data prediction to determine 97% correlation between concentration and crosslinking time under various conditions

PROJECTS/PUBLICATIONS

Projects/Patents | *EEG Vector Database, Opioid-Overdose Prevention Device*

- EEG Vector Database** - Used Wilson-Cowen model to generate E-I patterns and replicate EEGs of Parkinson's and healthy patients; used CEBRA package to transform EEGs into vector embeddings plotted into vector database; trained offset-10 model to identify differences between Parkinson's and healthy patient EEGs
- Opioid-Overdose Prevention Device (Provisional Patent- U.S. 63/657,693)** - Developed a modular Opioid-overdose prevention device which dispenses naloxone based on patients respiratory rate, heartbeat, and other physiological factors using a proprietary microneedle dispensing method; currently developing full-stack app

Publications/Presentations | *MRS Communications, MRS, Boston 2024, APS Summit 2025, JESTI 2025*

- Paper + Oral Presentation** - Machine Learning for Assessing Impact of Polydispersity on Thin Film Thickness
- Paper** - Ethics of ML-Aided Treatment Planning and Early Detection of Neurodegenerative Diseases

SKILLS

Technical: Python, Javascript, Ansible, CI/CD, SQL, R, PyTorch, AWS Lambda, Supabase, PostgreSQL, OOP, Rest APIs, Docker, Vercel, Pandas, Scikit-learn, Git, Linux, LangGraph, RAG, Async I/O, ChromaDB